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In [2]: # Program 18: PCA on E-Commerce Customer Behavior dataset (reduce to 2D)
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA

FILE = r"D:\Datascience-MachineLearning\Datasets\ecommerce_customer_behavior_dataset.csv" # <-- change to your file path

df = pd.read_csv(FILE)
X = df.select_dtypes(include="number").dropna() # numeric features only

X_scaled = StandardScaler().fit_transform(X)

pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)

print("Original shape :", X.shape)
print("Transformed shape:", X_pca.shape)
print("Variance explained by each component:", pca.explained_variance_ratio_)

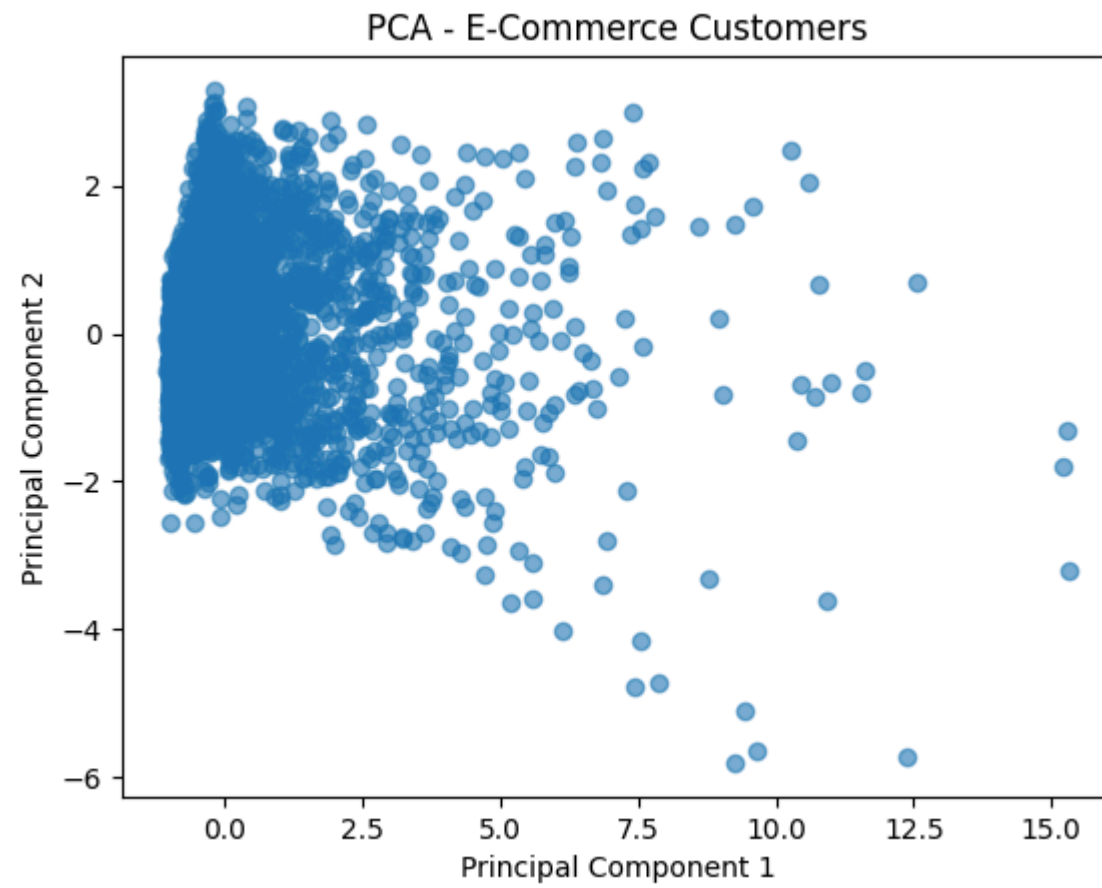
plt.scatter(X_pca[:, 0], X_pca[:, 1], alpha=0.6)
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("PCA - E-Commerce Customers")
plt.show()

```

Original shape : (5000, 9)

Transformed shape: (5000, 2)

Variance explained by each component: [0.23605761 0.11924352]



In []: